Method 3: angle equals the angle between the ray tangents at the observer:

From the ray equation , calculate :

Insert observer coordinates and from , noting small :

This agrees with , again giving

Method 4: use to rewrite as

Thus

For , the asymptote of is

This straight line has intercept and slope

Thus Substitute for to obtain , and hence

7.

Helium has two electrons; is the energy to remove one to infinity, satisfying . For , find the ion energy after one electron is ionized, : one electron orbiting the helium nucleus. Here are two methods.

Method 1: in the mechanical equation

is the radius and the speed. Bohr's ground-state quantization ( ) gives

Thus the helium-ion energy is